Joey Rudd

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Education

York University, Lassonde School of Engineering — Toronto, ON
Honours Bachelor of Computer Science — *Expected May 2028*

Experience

Software Developer (Co-op) — Jan 2026 – Dec 2026
Ontario Science Centre — *Toronto, ON*

- Deployed two interactive games as in-person exhibits at the Harbourfront Centre, reaching **5,000+ visitors** and logging **4,200+ game sessions** over three months
- Optimized exhibits for 4K displays, multi-touch kiosk input, and WCAG 2.1 AA accessibility standards
- Developed Extrema Buddies, a virtual pet game in Godot (GDScript) driven by state machines and physics
- Built Launch Into Rockets, a scroll-driven 3D explainer in Three.js with 3D models, parallax effects, and carousels covering each part of a rocket

Projects

Human Body Atlas | *Three.js, JavaScript, Vite*

- Built a 3D anatomy explorer at the Ontario Science Centre that lets users switch between five body systems (skeletal, digestive, cardiovascular, respiratory, nervous) with animated fade transitions
- Implemented raycast-based touchpoints with tweened camera zoom, loading each system's content from JSON
- Loaded a Meshopt-compressed glTF anatomy model and added full keyboard navigation across touchpoints and system controls for accessibility

AI Fitness Coaching App | *Python, FastAPI, Vue.js, Gemini API*

- Architected a full-stack coaching app with Vue 3 Composition API, async FastAPI, and Google Gemini, reaching a **100/100 Lighthouse** Performance score with automated Railway and Netlify deployment
- Developed a hybrid RAG system using BM25 indexing with TF-IDF and sentence-transformer fallbacks, retrieving context from **800+ knowledge chunks** to ground LLM responses
- Wrote a **57-test suite** covering chat flow, error handling, intent detection, and TDEE calculations, run through a GitHub Actions CI/CD pipeline

Science Bonds | *React, JavaScript*

- Built a word-matching science game in React at the Ontario Science Centre that records player test data
- Deployed the game offline at an Ontario Power Generation event, storing session data locally on site

Technical Skills

Languages: Java, Python, JavaScript, Go, C, SQL, GDScript, HTML/CSS
Frameworks: React, Vue.js, FastAPI, Spring, Three.js, JUnit, TailwindCSS
Developer Tools: Git, GitHub Actions, Docker, Godot, Vite, Unix/Linux, Netlify, Railway
Databases: SQLite